SYCAMORE LANE
AUTO DIAGNOSTICS
INDEPENDENT VEHICLE DIAGNOSTICS · SINCE 1998

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VEHICLE DIAGNOSTIC REPORT

Job no. SLA-26-119-44
Inspected Fri 29 May 2026
Workshop Bay 3, Hounslow

Independent diagnostic report prepared at the customer's request following an in-service failure. Issued without affiliation to any party to a sale.

Customer	Mr Marcus James Thompson
Address	32 Beechcroft Avenue, Twickenham TW2 5LA
Telephone	07733 401 882
Email	marcus.j.thompson@gmail.com
Booked	27 May 2026
Job ref.	SLA-26-119-44

Vehicle	BMW 320i M Sport (F30), 2018
Registration	LV68 KFE
VIN	WBA8E5C50KA192417
Engine	B48A20A - 2.0 L petrol turbo
Engine no.	B48A20A0192417
Mileage at inspection	47,996 miles

1. CUSTOMER'S REPORTED SYMPTOMS

The customer reports that, on Wed 27 May 2026, while travelling at approximately 65 mph on the M4 westbound, the engine management lamp illuminated and the vehicle entered limp-home mode (a noticeable loss of power, RPM held at approximately 3,000 rpm and a 'spool' rattle audible from the front of the engine). The customer pulled onto the hard shoulder, shut the engine down and called recovery. The vehicle has not been driven since and was delivered to this workshop by low loader. The customer states that the vehicle had been driven approximately 180 miles since collection from the selling dealer on 15 May 2026, with no prior warning lights or audible faults reported.

2. INITIAL VISUAL INSPECTION

The vehicle was received cold. Exterior, interior and underbonnet presentation is consistent with a recent detail clean. The following observations were recorded prior to fault-code scanning:

Engine bay	Recently cleaned, mild residual degreaser on the subframe. No fresh oil mist on the underside of the bonnet, suggesting the leak is dry or recent.
Oil level / quality	Marginally above MIN, dark grey, faint metallic sheen on the dipstick - suspicious of bearing wear.
Intercooler hose (charge side)	A film of engine oil (~2 ml) present in the intercooler-to-throttle hose. Typical of compressor-side seal failure.
Exhaust tailpipe	Light black sooting at the tip; intermittent over-fuelling consistent with closed-loop AFR compensation.
Boost hoses & clips	All clips present and torqued. No split hoses identified. No external boost-leak evident at idle (smoke-tested).
Air filter	OEM filter fitted, no obvious oil ingress on the clean side.

Service stamps / DSA	Last recorded service 23/11/2025 at 44,180 mi. No record of a turbo-related complaint on the DSA history.

3. DIAGNOSTIC FINDINGS (OBD-II SCAN)

Scan tool: Autel MaxiSYS MS909 (firmware 1.93). Communication established with DME, DSC, EGS and ZGW. Fault codes retrieved from the DME (engine control module):

Code	Description	Freeze frame (rpm / load / IAT)	Status
P0299	Turbocharger / supercharger underboost - actual boost below commanded by 280 mbar.	3,150 rpm / 78 % / 32 °C	Active
P02CC	Turbocharger boost-control 'A' position sensor - circuit intermittent.	2,820 rpm / 64 % / 31 °C	Active
P0234	Turbocharger / supercharger overboost - compensating actuator hunting.	4,180 rpm / 91 % / 36 °C	Stored
P0171	System too lean (Bank 1) - long-term fuel trim +18.5 % during the limp event.	2,510 rpm / 55 % / 29 °C	Stored
P0420	Catalyst efficiency below threshold (Bank 1) - consistent with prolonged rich/lean swing.	Idle / 18 % / 27 °C	Pending

Adaptation memory shows the long-term fuel trim has been progressively positive for the preceding ~1,400 miles of logged drive cycles, indicating the lean condition pre-dates the present owner's tenure.

4. LIVE DATA & BENCH TESTS

- Boost actuator stroke test: commanded 0–100 % via the scan tool. Actuator reaches only 78 % of commanded travel and intermittently sticks at the upper end of its range.
- Charge-air pressure log (road test): commanded boost held steady at 1.4 bar absolute; measured boost peaked at 1.18 bar and collapsed to 1.02 bar within 1.8 seconds. Repeatable on two separate runs.
- Shaft inspection: turbocharger feed pipe removed. Visible axial play of 0.32 mm and radial play of 0.18 mm at the compressor wheel (BMW service limit: 0.05 mm / 0.08 mm).
- Wastegate actuator: manual lever movement shows intermittent binding. Vacuum pump test confirms the actuator diaphragm holds but releases ~0.4 s slower than spec.
- Compressor seal: boroscope inspection of the intake side shows oil residue and a thin coating on the impeller hub - consistent with internal seal failure rather than a one-off blow-by event.
- Engine compression: dry test 14.2 / 14.1 / 14.3 / 14.2 bar across cylinders 1–4 (within spec). No indication of head-gasket or cylinder distress; failure is isolated to the turbocharger.
- Audible inspection (cold start): a low-frequency 'rumble' audible from the turbocharger housing at 2,400–2,800 rpm under light load - characteristic of bearing-cage wear.

5. PROBABLE CAUSE & TIMELINE ASSESSMENT

The combination of (a) progressive positive fuel trims in the adaptation memory dating back 1,400 mi, (b) physical shaft play well outside the manufacturer's service limit, (c) oil contamination of the charge-air tract and (d) audible bearing noise at moderate load is consistent with a turbocharger that had been deteriorating over a period of weeks to months prior to the failure event. A single-event failure in the 180 miles since the customer took delivery is not supported by the evidence - in particular, the long-term fuel trim history and the catalyst efficiency code (P0420 pending) both reflect behaviour that pre-dates the customer's tenure.

In my professional opinion, a competent pre-sale inspection (including a scan of the adaptation memory and a moderate-load test drive of more than 10 minutes) would have identified the audible bearing noise, the progressively positive fuel trims and the boost variation visible on live data. I do not consider this fault to have arisen suddenly during the 12-day period of customer ownership.

6. RECOMMENDATIONS

- Replace the turbocharger with an OE-grade unit and renew the oil-feed and return lines as a precaution.
- Carry out a full oil and filter service immediately following the turbo replacement to remove debris carried by the failed compressor seal.
- Inspect and, if necessary, replace the intercooler core and all charge-air pipework; oil deposits in the intercooler will otherwise be ingested into the new turbocharger.
- Renew the engine air filter and the cabin filter while access is available.
- Following repair, complete a 20-mile road test under a range of loads and clear adaptation memory before returning the vehicle to the customer.
- Retain all replaced parts for inspection in support of any subsequent claim under the Consumer Rights Act 2015.

7. ESTIMATE TO REPAIR

Item	Parts	Labour	Total
Replacement turbocharger (OE-grade, supplied)	£2,150.00	£680.00	£2,830.00
Oil-feed and oil-return pipes (renewal)	£135.00	£85.00	£220.00
Engine oil and filter service post-fit (BMW LL-17 FE+, 5W-30)	£95.00	£45.00	£140.00
Intercooler clean / charge-air pipework flush	£40.00	£95.00	£135.00
Air filter & cabin filter (renewal)	£58.00	£25.00	£83.00
Diagnostic re-check, adaptation reset, road test	included	£110.00	£110.00
Sundries (gaskets, fluids, fasteners)	£82.00	included	£82.00
Estimated subtotal (excluding VAT)			£3,600.00
VAT @ 20 %			£720.00
Estimated total (inc. VAT)			£4,320.00

This estimate is valid for 30 days from the date of inspection. Parts supplied by a third party are excluded from any workshop warranty. Workshop warranty on parts supplied and fitted: 24 months or 24,000 miles, whichever comes first.

8. TECHNICIAN'S DECLARATION

I confirm that the inspection and findings recorded above were carried out personally by me on the date and at the location shown. The report represents my professional opinion as a qualified diagnostic technician and is given independently of any party to the customer's purchase of the vehicle. I am willing, if required, to attend court or produce a witness statement in support of this report.

Signed by the diagnostic technician

D Khan

Daniyal Khan, IMI Adv. Dip. (Auto Diag.)
Senior Diagnostic Technician · ATA No. 081244-09
Date: 29 / 05 / 2026

Authorised on behalf of
Sycamore Lane Auto Diagnostics Ltd
Job ref. SLA-26-119-44
Workshop ticket and photographs retained on file

Report prepared at the customer's request. May be relied upon for the purpose of establishing the condition of the vehicle at the date of inspection.